

Comprehensive Lesson Plan: Introduction to Functions

Part 1: Lesson Plan for Graduate Portfolio

Teacher: Abraham Romero **Course:** 8th Grade Math **Grade Level:** 8th Grade Math **Topic:** Introduction to Functions: Defining, Representing, and Identifying **Date Implemented/Recorded:** Monday, December 1, 2025 **Time Allotment:** 90 minutes

I. Learning Goals / Objectives:

Upon completion of this lesson, students will be able to:

- Define a function as a special relationship where each input has exactly one output.
- Represent functions and relations using ordered pairs, tables, and mappings.
- Identify whether a given relation is a function using various representations (ordered pairs, tables, mappings, and graphs).
- Utilize technology to explore, visualize, and analyze functional relationships.

II. Texas Essential Knowledge and Skills (TEKS) / Standards:

- **8.5(G):** The student is expected to identify functions using sets of ordered pairs, tables, mappings, and graphs.
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III. Materials:

- Teacher: Computer with internet access, projector, [specific classroom setup for recording, if applicable, e.g., external microphone, tripod]
 - Students: Individual computing devices (Laptops/Chromebooks/Tablets) with internet access
 - Platform: **Desmos Activity Builder via Amplify Curriculum**
 - Assessment Tool: Google Forms (for Exit Ticket)
 - Student Supplies: Paper and pencil for independent work and reflection.
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IV. Assessment (Pre & Post):

- **Pre-Assessment (Formative - Integrated in Engage Phase):**

- The interactive Function Machine activity within Desmos from Amplify serves as a dynamic pre-assessment. By observing students' initial drag-and-drop interactions, their recorded output predictions, and the automatically generated table, the teacher can gauge their intuitive understanding of input-output relationships and identify existing knowledge or misconceptions regarding consistent outputs. This real-time data informs subsequent instructional adjustments.
 - **Post-Assessment (Summative - Evaluate Phase):**
 - **Exit Ticket (via Google Form):** Students will complete a short, digital exit ticket linked through Desmos, specifically asking them to determine whether given relations (presented as sets of ordered pairs or mappings) represent functions. This directly assesses their individual mastery of the lesson's core definition and identification skills.
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V. Technology Integration:

Technology is integral to the delivery and success of this lesson, fostering an interactive and visually rich learning experience that enhances understanding of abstract algebraic concepts.

- **Desmos Activity Builder (via Amplify Curriculum):** This platform is strategically utilized for:
 - **Interactive Conceptualization:** The interactive Function Machine allows students to actively engage with input/output rules by manipulating numerical blocks and immediately observing corresponding outputs and graphical transformations. This hands-on approach makes abstract ideas concrete and provides instant visual feedback.
 - **Pacing and Differentiation:** The teacher effectively paces student progress through the Desmos activities, providing individualized support or challenges as needed, directly observable through the teacher dashboard.
 - **Real-time Formative Assessment:** The teacher continuously monitors live student input and work on custom screens within Desmos (e.g., digital whiteboards), enabling immediate feedback, identification of common misconceptions, and data-driven guidance for whole-class or small-group discussions. The platform facilitates showcasing and analyzing anonymous student work.
 - **Dynamic Representations:** Desmos automatically generates tables of inputs and outputs as students interact, actively supporting the development of varied functional representations.
- **Google Forms:** Provides a streamlined method for deploying a quick, trackable digital exit ticket, offering concrete data for summative evaluation.

VI. Lesson Procedure (5E Instructional Model):

(A) Engage (Approx. 15 minutes)

- **Purpose:** To capture student interest, connect to the concept of rules transforming numbers, and activate initial understanding of input-output relationships, setting the foundation for functions.
- **Teacher Activity:**
 - Direct students to the Desmos activity, launching the interactive Function Machine.
 - Prompt students to experiment with inputs, encouraging them to observe the outputs and generated tables.
 - Guide the class to consider what rules might be "inside the machine" and to hypothesize functions based on their observations.
 - Foster initial discussion about the consistent relationship between inputs and outputs.
- **Student Activity:**
 - Access and independently interact with the virtual Desmos Function Machine, experimenting with various input blocks.
 - Analyze the changing outputs and dynamic table automatically generated by Desmos.
 - Formulate hypotheses about the function rule in groups or individually.
 - Engage in initial discussions, sharing observations and predictions.

(B) Explore (Approx. 25 minutes)

- **Purpose:** Students will actively investigate the linear relationships within function machines, translating observations into equations while bridging to prior knowledge of slope and y-intercept. This phase intentionally addresses common misconceptions identified from recent assessments.
- **Teacher Activity:**
 - Present a new set of input-output pairs (e.g., from a Desmos screen or projector) that represents a linear function.
 - Guide students to select two points and apply the slope formula, encouraging them to use paper and pencil for formal calculation.
 - **Strategically address common errors from the previous test:** Emphasize the importance of explicitly writing down the slope formula and correctly handling negative values in calculations, drawing direct connections to prior misconceptions.
 - Facilitate students in calculating the rate of change (slope) and determining the y-intercept through careful questioning and prompting.

- Lead the class to articulate the equation of the function discovered inside the machine.
- **Student Activity:**
 - Select two input-output pairs from the given data.
 - Independently or collaboratively (as directed) apply the slope formula using paper and pencil, demonstrating careful notation and calculation.
 - Determine the rate of change (slope) and y-intercept based on their calculations.
 - Formulate the equation of the linear function.
 - Verify results with peers and participate in class discussion to confirm understanding.

(C) Explain (Approx. 15 minutes)

- **Purpose:** To formalize student understanding by providing clear definitions, introducing mathematical notation for relations and functions, and explicitly connecting functional concepts to various representations.
- **Teacher Activity:**
 - Present a new Desmos screen or use the projector to introduce the concepts of domain (set of inputs) and range (set of outputs) using examples (e.g., inputs {2,4,6} leading to outputs {1,2,3}).
 - Guide students to identify the function rule for this new example through class discussion.
 - Formally introduce the visualization of a **relation** as a **mapping** (arrows from inputs to outputs).
 - Model the conversion of mappings into a **set of ordered pairs**, emphasizing correct brace notation ($\{ \}$).
 - Formally define a **function** as a relation where "each input has **one and only one** output," clarifying that inputs are x-values and outputs are y-values.
 - Check for understanding by asking students to rephrase the definition or provide simple examples.
- **Student Activity:**
 - Participate in defining and identifying inputs (domain) and outputs (range) for given sets.
 - Observe and interpret mappings between sets, translating them into ordered pairs.
 - Take notes on formal definitions (relation, mapping, function, input, output, ordered pair, set notation).
 - Actively contribute to class discussions, articulate definitions, and ask clarifying questions.

(D) Elaborate (Approx. 20 minutes)

- **Purpose:** Students will apply their formalized understanding of function definitions and representations to analyze various relations and confidently distinguish functions from non-functions.
- **Teacher Activity:**
 - Transition to a series of challenge problems presented via Desmos or projector, showcasing various relations (sets of ordered pairs, mappings).
 - Direct students to independently or collaboratively (as guided) determine if each presented relation is a function.
 - Prompt students to justify their reasoning by explicitly referencing the "one input, one output" rule.
 - **Focus on critical comparative examples:** Present a relation that *is* a function (e.g., $\{(5, 8), (11, 9), (6, 8), (8, 5)\}$) and a relation that *is not* a function (e.g., $\{(2, 3), (2, 4), (2, 5), (2, 6)\}$ or repeating inputs like $(1, 7), (1, 2)$).
 - Facilitate a class discussion to review answers, specifically highlighting why non-functions fail the "one input, one output" criterion.
 - Address any lingering misconceptions and reinforce the connection between representations and the definition.
- **Student Activity:**
 - Analyze provided relations (ordered pairs, mappings, tables) within Desmos or on paper.
 - Apply the formal definition of a function to each relation to determine if it is a function.
 - Justify their conclusions, articulating which inputs have multiple outputs if a relation is not a function.
 - Participate actively in class discussions, sharing their reasoning and evaluating peers' responses.

(E) Evaluate (Approx. 15 minutes)

- **Purpose:** To formally assess individual student mastery of defining and identifying functions from various representations.
- **Teacher Activity:**
 - Instruct students to access and complete the Google Form Exit Ticket, sharing the link via Desmos.
 - Review instructions for the exit ticket, clarifying that questions require them to identify functions based on the lesson's criteria.
 - Circulate to monitor student progress and provide support for technical access if needed.
 - Collect devices or ensure form submission after completion.
- **Student Activity:**
 - Independently access and complete the Google Form Exit Ticket on their devices.

- Answer questions assessing their ability to identify whether specific relations represent functions, demonstrating their individual mastery of the lesson's objectives.
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